BIO-301 SUBJECTIVE MIDTERM-100% GUESS PAPER 2023

Essentials of Genetics

BY SULMAN ALI

0341-7547044

SUBJECTIVES

Note on oligonucleotide in mapping of disease gene.

Oligonucleotide is a short DNA sequence that can be used in mapping disease genes. These sequences can be used as probes to detect the presence of disease genes in a sample. Oligonucleotide probes can be synthesized to match the sequence of a specific gene, and then used to detect the presence of that gene in a sample. This technique is useful in identifying the genetic basis of diseases.

Genetic and environmental causes of variation.

Genetic causes of variation include mutations and genetic recombination, difference in DNA sequence that can affect traits such as eye color, hair color, and susceptibility to diseases.

Environmental variation: Differences in traits that are caused by environmental factors such as nutrition, exposure to toxins, and access to healthcare. while environmental causes of variation include exposure to different environmental factors such as temperature, light, and diet.

Spermatogesis and oogenesis.

Spermatogenesis is the process of producing sperm, while **oogenesis** is the process of producing eggs. Spermatogenesis occurs in the testes of males, while oogenesis occurs in the ovaries of females.

Difference between prokaryotes and eukaryotes.

Prokaryotes are unicellular organisms that do not have a nucleus or membrane-bound organelles, while **eukaryotes** are multicellular organisms that have a nucleus and membrane-bound organelles.

Common Epigenetics regulated phenomena:

Common epigenetic phenomena include DNA methylation, histone modification, and non-coding RNA regulation.

CEN DNA.

CEN DNA is a sequence of DNA found in eukaryotic chromosomes that is responsible for attaching the chromosomes to spindle fibers during cell division.

CAUSES OF EPIGENETICS:

3= DNA methylation, 2= Histone modification, 3= miRNAs

Number of chromosomes in chicken, house fly, gold fish?

Chicken has =78 chromosomes, house fly has= 12 chromosomes, and goldfish has =94 chromosomes.

Enlist the genes in drosophila of linkage mapping.

The genes in Drosophila of linkage mapping include white, yellow, brown, and ebony.

Explain the conformation of DNA.

The conformation of DNA is a double helix structure composed of nucleotides. The nucleotides consist of a sugar molecule, a phosphate group, and a nitrogenous base. The nitrogenous bases include adenine, thymine, cytosine, and guanine. Adenine pairs with thymine and cytosine pairs with guanine through hydrogen bonds. The double helix structure of DNA is stabilized by hydrogen bonds between the base pairs and the hydrophobic interactions between the stacked bases.

Note on staining procedures.

Staining procedures are used in microscopy to enhance the contrast and visibility of biological specimens. Different types of stains can be used to highlight different structures within a sample. For example, hematoxylin and eosin stains are commonly used to differentiate between different types of cells and tissues. Gram staining is used to differentiate between different types of bacteria. Staining procedures can also be used to detect the presence of specific molecules within a sample, such as antibodies or enzyme.

What type of variation occur in human:

1= Variations in skin color 2= Variation in hair color 3= Variation in hair curliness 4= Variation in eye color 5= Variation in sex.

Difference btw continuous and discontinuous variation.

Discontinuous variation entirely genetically controlled. They cannot be altered by external condition. blood group, color blind and dwarfism are genetically controlled.

Continuous variation describe the situation in which there are many intermediate between the extremes. Example shade of hair color between the black and blond.

Define Division of meiosis:

Meiosis is a cellular division process that produces haploid cells with half the chromosome number of the parent cell. It involves two rounds of division, resulting in four genetically diverse daughter cells.

What is meaning of exception of mendelian inheritance?

The exceptions to Mendelian inheritance are patterns of inheritance that do not follow the rules of dominance, segregation, and independent assortment described by Gregor Mendel. Some examples of exceptions to Mendelian inheritance include: 1= Incomplete dominance: 2= Codominance: 3= Multiple alleles: 4= Polygenic inheritance: 5= Epistasis:

Define leading strand, Lagging strand, okazaki fragments. Gyrase and sister chromatids

In DNA replication, the **leading strand** is synthesized continuously in the 5' to 3' direction, following the replication fork, while the **lagging strand** is synthesized discontinuously in short fragments called Okazaki fragments. **Okazaki fragments** are later joined by DNA ligase to form a continuous strand. **Gyrase** is an enzyme that helps relieve the tension caused by DNA unwinding during replication. **Sister chromatids** are two identical copies of a chromosome

joined together at the centromere, formed during DNA replication, and are separated during cell division.

Write note on genome imprinting:

Genomic imprinting is an epigenetic mechanism where certain genes are expressed differently based on their parental origin. Imprinted genes are marked with epigenetic tags, such as DNA methylation, during gamete formation. This leads to parent-specific gene expression patterns that can influence embryonic development, growth, and disease susceptibility. Imprinting disorders can result from abnormal gene expression patterns and have significant implications for human health.

What happens in anaphase:

In anaphase, the sister chromatids of each chromosome separate and move towards opposite poles of the cell. Microtubules attached to the centromeres shorten, pulling the chromatids apart and ensuring equal distribution of genetic material.

Role of positive and negative apistasis:

Positive epistasis occurs when the effect of one gene enhances the effect of another gene, leading to a more pronounced phenotype. Negative epistasis, on the other hand, occurs when the effect of one gene suppresses or diminishes the effect of another gene, resulting in a reduced phenotype.

Define nucleotide, nucleoside, purine, and pyrimidine.

A nucleotide is a building block of nucleic acids (DNA and RNA) consisting of three components: a sugar (ribose or deoxyribose), a phosphate group, and a nitrogenous base. A **nucleoside** is a similar structure, but lacks the phosphate group. **Purines** (adenine and guanine) and **pyrimidines** (cytosine, thymine, and uracil) are the two types of nitrogenous bases. Purines have a double-ring structure, while pyrimidines have a single-ring structure. Nucleotides combine to form the backbone of DNA and RNA strands.

Name five mapping technique.

1= Genetic mapping 2= Physical mapping 3= Radiation hybrid mapping 4= Comparative mapping 5= Restriction fragment length polymorphism (RFLP) mapping.

Define Classical and reverse genetics:

Classical genetics refers to the study of phenotypic traits and inheritance patterns to understand gene function. It involves crossing organisms with specific traits to analyze their inheritance. **Reverse genetics** is a more modern approach to studying genetics that involves starting with a known gene and then determining its function by creating mutations in the gene and observing the resulting phenotype. This approach is useful for studying the function of specific genes and for identifying potential drug targets.

Define Interphase, particulate and blending hypothesis, mono allelic, diallelic and poly allelic.

Interphase is the longest phase of the cell cycle, where the cell grows, carries out metabolic activities, and duplicates its DNA in preparation for cell division.

The particulate hypothesis suggests that hereditary traits are discrete particles (genes) passed from parents to offspring, while the **blending hypothesis** proposes that traits blend in offspring.

Monoallelic refers to a gene having only one active allele, while **diallelic** indicates the presence of two alleles. **Polymorphism** occurs when a gene has **multiple alleles** in a population, resulting in genetic diversity.

Describe the Prophase 1:

Prophase I is the first phase of meiosis I. During this stage, chromatin condenses into chromosomes, homologous chromosomes pair up in a process called synapsis, and crossing over

occurs. The nuclear envelope breaks down, and spindle fibers start to form, preparing for the subsequent stages of meiosis.

Explan in dihybride percentage give to rryy in the cross b/w RRYY and rryy:

In the cross between RRYY and rryy, all the offspring will be RrYy (heterozygous for both traits). Therefore, in the dihybrid cross, the percentage of offspring with the genotype rryy (both recessive traits) would be 0%.

Note on lambrush chromosomes:

Lampbrush chromosomes are large, highly extended chromosomes that are found in the oocytes (egg cells) of some animals, including birds, reptiles, and amphibians. They are named for their brush-like appearance, which is due to the presence of thousands of loops of chromatin that are attached to a central axis. Overall, lampbrush chromosomes are a unique and fascinating feature of the animal kingdom, and have played an important role in our understanding of gene expression and genome organization.

Universal donor and recipient.

The universal donor is type O negative blood, as it lacks A, B, and Rh antigens, making it compatible with all blood types. The universal recipient is type AB positive blood, as it can receive blood from all types since it lacks antibodies against A, B, and Rh antigens.

Give 4 steps of telophase:

1= Chromosomes start to uncoil. 2= Nuclear envelope forms around the chromosomes at each pole of the cell. 3= Spindle fibers break down and dissolve. 4= Daughter nucleus is formed in each of the daughter cell.

Write some therapies of epigenetics:

Epigenetic therapies include DNA methylation inhibitors (e.g., azacitidine), histone deacetylase inhibitors (e.g., vorinostat), and histone methyltransferase inhibitors (e.g., chaetocin). These drugs target epigenetic modifications to regulate gene expression and have potential applications in cancer treatment, neurological disorders, and other diseases.

Compare chromosome with chromatin:

Chromosomes are condensed structures composed of DNA, proteins, and other molecules. They are visible during cell division and carry genetic information. **Chromatin**, on the other hand, refers to the relaxed, uncondensed form of DNA and associated proteins. It exists during interphase and allows for gene expression and DNA replication.

Haploid diploid sex determination:

No sex chromosomes in most species of ants and bees. Females fertilized ova diploid. Males unfertilized ova haploid. So Different organism have different systems of sex determination. Haploid sex determination refers to the determination of an individual's sex based on the presence of a single set of chromosomes (n). Diploid sex determination involves the determination of sex based on the presence of two sets of chromosomes (2n).

Chargaff first second party rule also define GC rule:

FIRST PARITY RULE: Stated that purines pair with pyrimidines, specifically A with T and C with G

SECOND PARITY RULE: the % of A approximately equals % of T and the % of G approximately equals % of C.

G-C RULE: Stated that (G+C) % is constant within a species, but often differs between species.

Pleiotropic affect with three example.

Pleiotropy is the phenomenon in which a single gene affects multiple traits. Here are three examples of pleiotropic effects: 1= Sickle cell anemia 2= Marfan syndrome: 3= Phenylketonuria (PKU).

Specific properties of meiosis I and meiosis II

Meiosis I is the first division of meiosis, where homologous chromosomes pair up, undergo crossing over, and separate, resulting in two haploid daughter cells. It involves reduction division and contributes to genetic diversity.

Meiosis II is the second division, similar to mitosis, where sister chromatids separate, resulting in four haploid daughter cells. It involves equational division and ensures the distribution of genetic material to the daughter cells.

Prophase 2 in meiosis

Prophase II is the initial phase of meiosis II. During this stage, the nuclear envelope disintegrates, and spindle fibers reassemble. Chromosomes, each consisting of two chromatids, become visible, and homologous chromosomes do not pair as in Prophase I.

Feed backs of Mendel's Law of Independent Assortments:

Mendel's Law of Independent Assortment states that different pairs of genes segregate independently during the formation of gametes. This law laid the foundation for understanding genetic inheritance and provided evidence for the existence of discrete units of inheritance, which later became known as genes.

Linkage mapping

Linkage mapping is a technique used to determine the relative positions of genes on a chromosome. It is based on the principle of genetic linkage, which is the tendency of genes that are physically close together on a chromosome to be inherited together.

How genetics help us:

Genetics helps us understand the inheritance of traits, diagnose and treat genetic disorders, predict the risk of certain diseases, develop personalized medicine, improve agricultural practices, and advance our knowledge of evolution and biodiversity.

Describe percentage of monoallelic, diallelic and polyallelic conditions in human.

Human **monoallelic** expression may be several times more common than the previous estimate of approximately 1%. **Diallelic** indels comprise approximately 8% of all human polymorphisms. **Polyallelic** system are less common in human genetics.

Write down the method that converts euchromatin to heterochromatin:

One method that converts euchromatin to heterochromatin is through histone modifications, specifically histone methylation and acetylation. These modifications can alter the chromatin structure, making it more condensed and less accessible for gene expression.

Explain co-dominance:

Co-dominance is a genetic phenomenon where both alleles of a gene are expressed simultaneously in the heterozygous state. Unlike dominance, where one allele masks the expression of another, co-dominance results in a combined phenotype that showcases the traits of both alleles. For example, in the ABO blood group system, individuals with the genotype AB exhibit both A and B antigens on their red blood cells, demonstrating co-dominance.

comparison b/w mitosis/ meiosis.

Mitosis:

1. One division
2. Produces two identical diploid cells

3. Maintains the same chromosome number
4. Limited genetic variation
5. Occurs in somatic cells
6. Purpose is cell replication, growth, and tissue repair

Meiosis:

1. Two divisions (meiosis I and II)
2. Produces four genetically diverse haploid cells
3. Reduces chromosome number by half
4. Significant genetic variation through crossing over and independent assortment
5. Occurs in reproductive cells
6. Purpose is the production of gametes for sexual reproduction.

Enlist 10 species chromosomes numbers:

1. Human (Homo sapiens): 46 (23 pairs)
2. Fruit fly (Drosophila melanogaster): 8
3. House mouse (Mus musculus): 40
4. Domestic dog (Canis lupus familiaris): 78
5. Indian muntjac (Muntiacus muntjak): 7
6. Garden pea (Pisum sativum): 14
7. Corn (Zea mays): 20
8. Tomato (Solanum lycopersicum): 24
9. African elephant (Loxodonta africana): 56
10. Wheat (Triticum aestivum): 42

Incomplete Dominance:

Incomplete dominance is a genetic inheritance pattern where neither allele is completely dominant over the other. Instead, the heterozygous phenotype is an intermediate blend of the two alleles, resulting in a distinct phenotype different from either homozygous genotype.

CEN-DNA and it's subunits:

CEN-DNA, or centromeric DNA, is a specialized DNA sequence found at the center of a chromosome's constriction point called the centromere. It serves as the binding site for proteins, including **subunits** such as CENP-A, CENP-B, and CENP-C, which are involved in proper chromosome segregation during cell division.

Sex linked traits and diseases in humans:

Sex-linked traits and **diseases in humans** are those that are associated with genes located on the sex chromosomes (X and Y). These traits and diseases are often inherited in a sex-dependent manner and can be more commonly observed in one sex over the other due to differences in chromosome inheritance.

Mandelian and non mandelian genetics:

Mendelian genetics refers to the principles of inheritance proposed by Gregor Mendel, involving the transmission of traits from parents to offspring through discrete units (genes) following predictable patterns. **Non-Mendelian genetics** refers to inheritance patterns that do not strictly adhere to Mendel's laws, including incomplete dominance, co-dominance, polygenic inheritance, and epistasis.

Is there any difference between telophase 1 and teleophase 2 of meiosis? please discuss:

Yes, there is a difference between telophase I and telophase II of meiosis. Telophase I follows the first division of meiosis and involves the separation of homologous chromosomes, while telophase II follows the second division and involves the separation of sister chromatids.

Define sex chromosome and define homogamatic and hetero gametic:

A sex chromosome is a type of chromosome that determines an individual's sex. In humans, the sex chromosomes are X and Y. Females typically have two X chromosomes (XX), while males have one X and one Y chromosome (XY).

Homogametic refers to individuals with identical sex chromosomes, such as females (XX). **Heterogametic** refers to individuals with different sex chromosomes, such as males (XY).

Write steps of map up bottom technique:

1= Make random clones, 2= Sequence each clone, 3= Overlap sequence reads, 4= overlap contings for complete sequence .

write the use of telomeres:

Telomeres play a crucial role in preserving the integrity and stability of chromosomes. They protect the ends of chromosomes from degradation, prevent fusion with other chromosomes, and help maintain the genomic stability during DNA replication and cell division.

How genetic mapping is formed?

Genetic mapping is formed by studying the inheritance patterns of genetic markers, such as genes or DNA sequences, in families or populations. By analyzing the frequency of recombination events, the relative positions of markers on a chromosome can be determined, creating a genetic map.

What is the difference between spermatogenesis and oogenesis?

Spermatogenesis:

1. Occurs in the testes.
2. Results in the production of four functional sperm cells.
3. Begins at puberty and continues throughout the male's life.
4. Produces haploid sperm cells.
5. Meiosis divisions are continuous, resulting in minimal cytoplasmic differentiation.
6. Produces millions of sperm cells daily.

Oogenesis:

1. Occurs in the ovaries.
2. Results in the production of one functional egg cell (ovum) and three polar bodies.
3. Begins before birth and ceases during menopause.
4. Produces haploid egg cells.
5. Meiosis divisions are discontinuous, with extensive cytoplasmic differentiation.
6. Produces one egg cell per month (in fertile women).

How many numbers of genotype produce in rabbits for fur color write the numbers of genotype with special color of the fur?

Rabbits have four possible genotypes for fur color: CC (Chinchilla), Cc (Himalayan), cchc (Himalayan with the extension gene), and cch (Himalayan with chinchilla dilution). These represent the major genotypes associated with specific fur color variations in rabbits.

EXCEPTIONS TO MENDELIAN INHERITANCE:

1= Incomplete dominance 2= Codominance 3= Multiple alleles 4= Polygenic traits 5=Epistasis 6= Pleiotropy 7= Environmental effects on gene expression 8= Linkage 9= sex linkage.

Difference between haploid cells and diploid cells:

Haploid cells have a single set of chromosomes (n), while diploid cells have two sets of chromosomes (2n). Haploid cells are typically gametes (sperm and eggs), while diploid cells are found in most somatic cells of the body.